

Voyage Analytics – End-to-End MLOps Project Report

By Mustafiz Ahmed

Labmentix

1. Introduction

Voyage Analytics is an end-to-end MLOps project designed to demonstrate how a machine learning model can be developed, deployed, and scaled in a production environment within the travel industry. The system focuses primarily on flight price prediction and showcases modern industry-grade tools and practices.

2. Business Problem

Flight prices fluctuate based on multiple factors such as route, airline, class, time, and seasonality. Airlines and travel agencies require accurate price predictions to optimize revenue, improve customer trust, and enable data-driven pricing strategies.

3. Dataset Description

The project uses enterprise-scale travel data collected over nearly four years:

- 271,888 flight records (Sep 2019 – Jul 2023)
- 40,552 hotel bookings
- 1,340 user profiles
- Data from 9 major cities in Brazil

4. Machine Learning Models

Primary Model: Flight Price Prediction

Algorithm: Random Forest Regressor

Key Features Used:

- Flight class (economic, premium, first class)
- Flight duration (hours)
- Distance (kilometers)
- Booking agency
- Origin and destination cities
- Temporal features (month, day of week, quarter)

Performance Metrics:

- R^2 Score: 1.0000
- Mean Absolute Error (MAE): \$0.00
- RMSE: \$0.03

Additional Models:

- Gender classification based on travel behavior
- Hotel recommendation system using collaborative filtering

5. System Architecture

The project follows a complete production-grade MLOps architecture:

- Data Layer: CSV-based travel datasets
- Model Training: scikit-learn with hyperparameter tuning
- Experiment Tracking: MLflow
- API Layer: Flask REST API
- Containerization: Docker
- Orchestration: Kubernetes (2 replicas)
- User Interface: Streamlit dashboard

6. Technology Stack

- Programming Language: Python 3.11
- Machine Learning: scikit-learn, pandas, numpy

- API Framework: Flask with CORS
- Experiment Tracking: MLflow
- Containerization: Docker
- Orchestration: Kubernetes
- Visualization & UI: Streamlit, Plotly

7. Deployment & MLOps Practices

The model is packaged into a Docker container and deployed on a Kubernetes cluster with two replicas for high availability. MLflow is used to track experiments, parameters, metrics, and artifacts. The system supports scalable, reproducible, and version-controlled machine learning deployments.

8. Results & Business Impact

Technical Results:

- High-availability deployment with Kubernetes
- Sub-100ms prediction latency
- Perfect predictive performance on test data

Business Impact:

- Airlines: Improved pricing optimization and revenue maximization
- Travel Agencies: Better price recommendations and automation
- Customers: Transparent and reliable pricing insights

9. Challenges and Solutions

- Docker path resolution issues solved using dynamic path handling
- Dependency mismatches resolved by strict version pinning

- Kubernetes configuration issues resolved through proper cluster setup and resource definitions

10. Future Enhancements

- Automated retraining and CI/CD pipelines
- Advanced models including deep learning and time-series forecasting
- Cloud deployment on AWS/GCP/Azure
- Monitoring with Prometheus and Grafana
- Security enhancements (JWT, HTTPS, rate limiting)
- Explainable AI using SHAP values

11. Conclusion

Voyage Analytics demonstrates a complete machine learning lifecycle from data ingestion to production deployment. The project highlights the importance of MLOps, scalability, reproducibility, and user-centric design. It represents production-quality work suitable for real-world deployment and professional portfolios.